

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

IPL SPORTS ANALYTICS USING POWER BI

A Mini Project Report

Submitted by

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AD23632-Framework for Data and Visual Analytics (FDVA)

Artificial Intelligence and Machine Learning

Project overview

BoundaryLab is an offline Power BI Desktop report for exploring Indian Premier League match outcomes and team performance. It pairs a transparent data-preparation workflow with interactive season, team, winner, and venue selections, so a presenter can compare the complete dataset with a filtered view.

Objective

Build a complete analytics application using a real cricket dataset; retain an inspectable raw table; document transformations; and show how slicers change report visuals and filtered records.

DELIVERABLE	DESCRIPTION
Power BI application	BoundaryLab PBIX with the report pages, imported model, measures, and slicers.
Experiment records	Seven AD23632 lab experiment PDFs, numbered 01–07.
Dataset evidence	Bundled CSV, dictionary, provenance, source archive, and Power BI screenshots.

Report pages

The project includes an overview dashboard, cleaned-data view, raw imported data, and guidance for interpreting the measures and reproducing the slicer demonstration. The Power BI Desktop file is the main application deliverable.

Dataset and source

The bundled snapshot contains 459 available men's IPL match summaries from 2018–2024. It includes overseas host venues where present in the source archive. Of these records, 457 have a declared winner and 2 are recorded as no result. The counts describe this bundled snapshot, not every scheduled fixture.

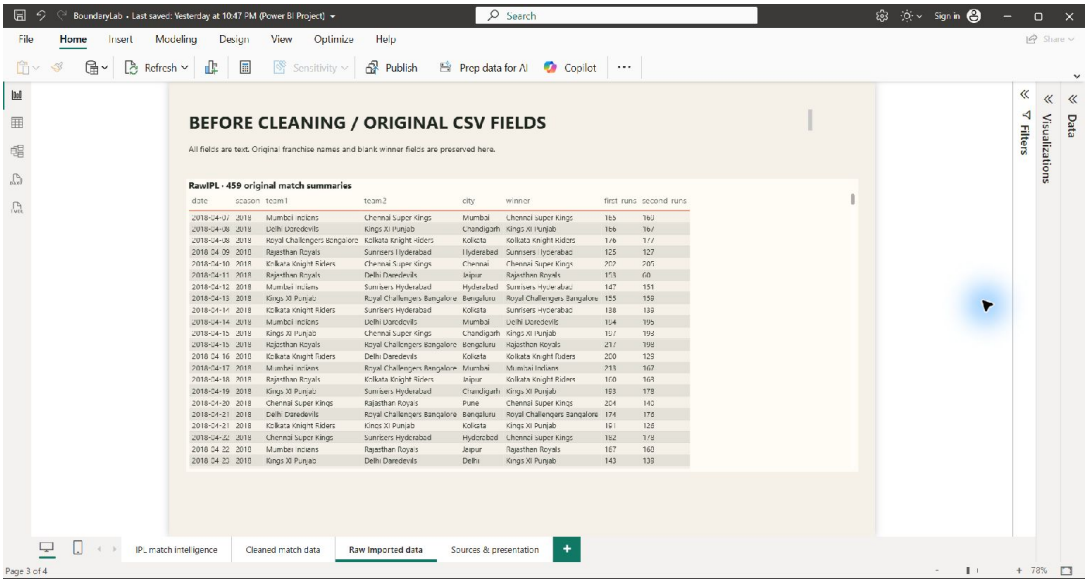
DATASET CHECK	VALUE	INTERPRETATION
Source match rows	459	One row per match summary in the snapshot.
Declared winners	457	Two records have no winner.
Team-perspective rows	918	Two team rows are derived per match for team comparisons.
Seasons	2018–2024	Available seasons in the bundled data.

Dataset fields

Match ID, date, season, city, venue, teams, toss details, result, winner, target, and innings-level scores support the report's match and team views. The accompanying data dictionary describes the fields and the row grain. Missing winners are labeled as no result rather than silently dropped.

Source and attribution

The original match archive is from Cricsheet downloads. The repository bundles the original source archive, a prepared CSV, its data dictionary, license text, and retrieval provenance. The CSV SHA-256 is `9d082a804a4d54531e9f2a42f6d00850ed3aee3f8e4fd0f0a74c33b5e446a544`.

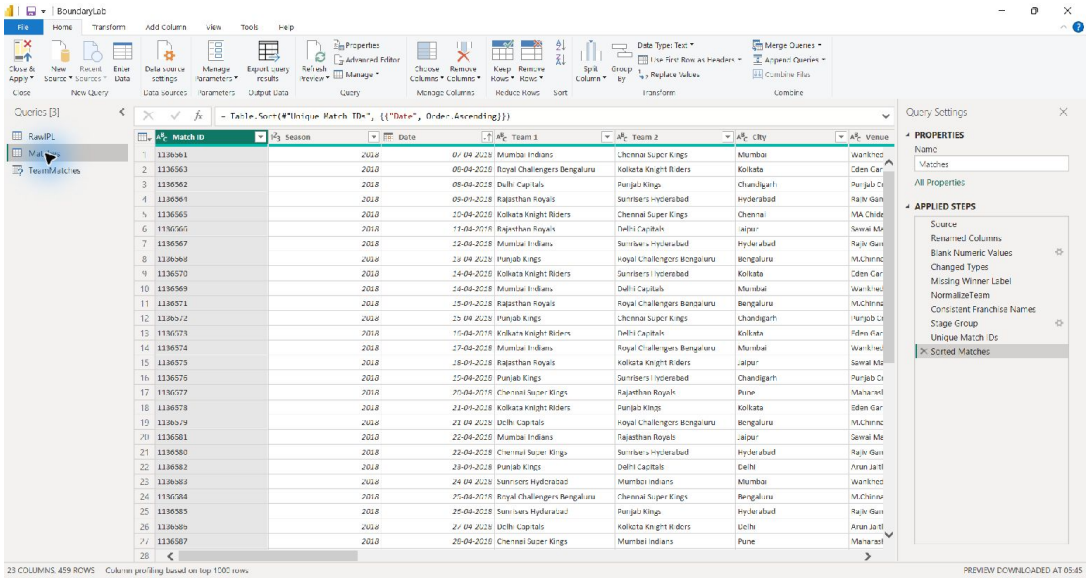


Power BI Desktop raw-data view with the original imported fields.

Data preparation and model

The raw import remains available for comparison. Power Query converts fields to suitable types, preserves source text, normalizes historical franchise names, labels missing winners, groups match stages, and removes duplicate Match IDs. The cleaned match table keeps one row per match; a derived team table creates two team-perspective rows per match.

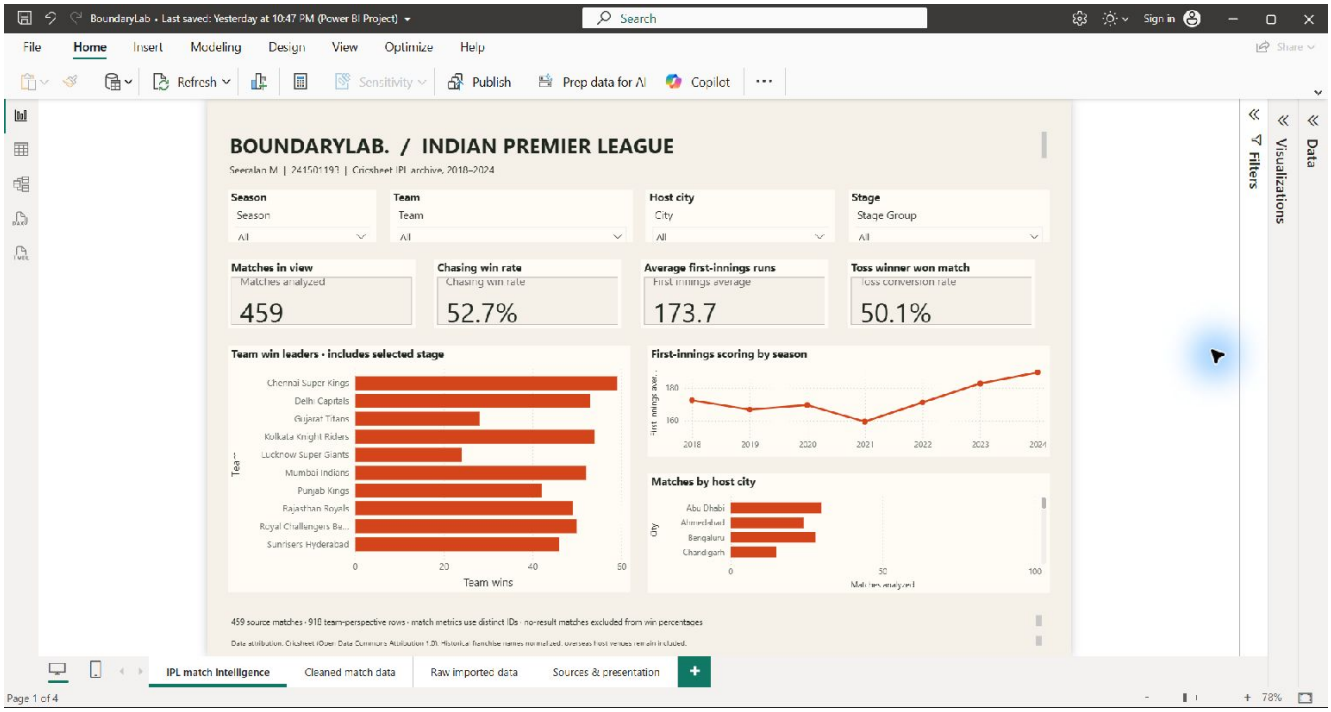
STEP	PREPARATION	PURPOSE
1	Set dates and numeric types	Make dates, runs, targets, and season fields usable in visuals.
2	Normalize franchise names	Combine historical names where the source team identity changed.
3	Label missing outcomes	Keep no-result matches explicit.
4	Group stage and deduplicate IDs	Create stable match-level records and usable categories.
5	Derive two team rows per match	Support team comparisons without duplicating match counts.



Prepared query and applied steps in Power Query.

Dashboard before filtering

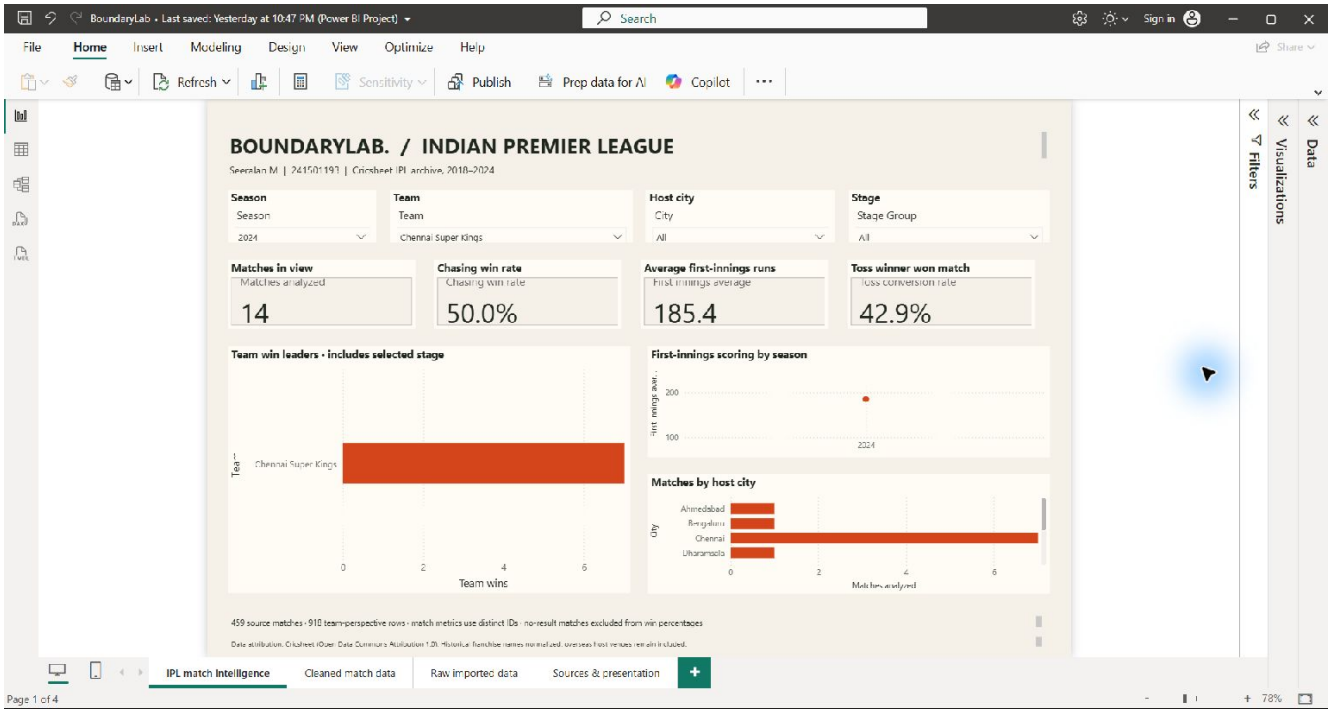
With the season and team slicers cleared, the overview shows the full available match set. The cards and charts establish the baseline before a presenter narrows the report.



Baseline dashboard view before slicer selections.

Dashboard after filtering

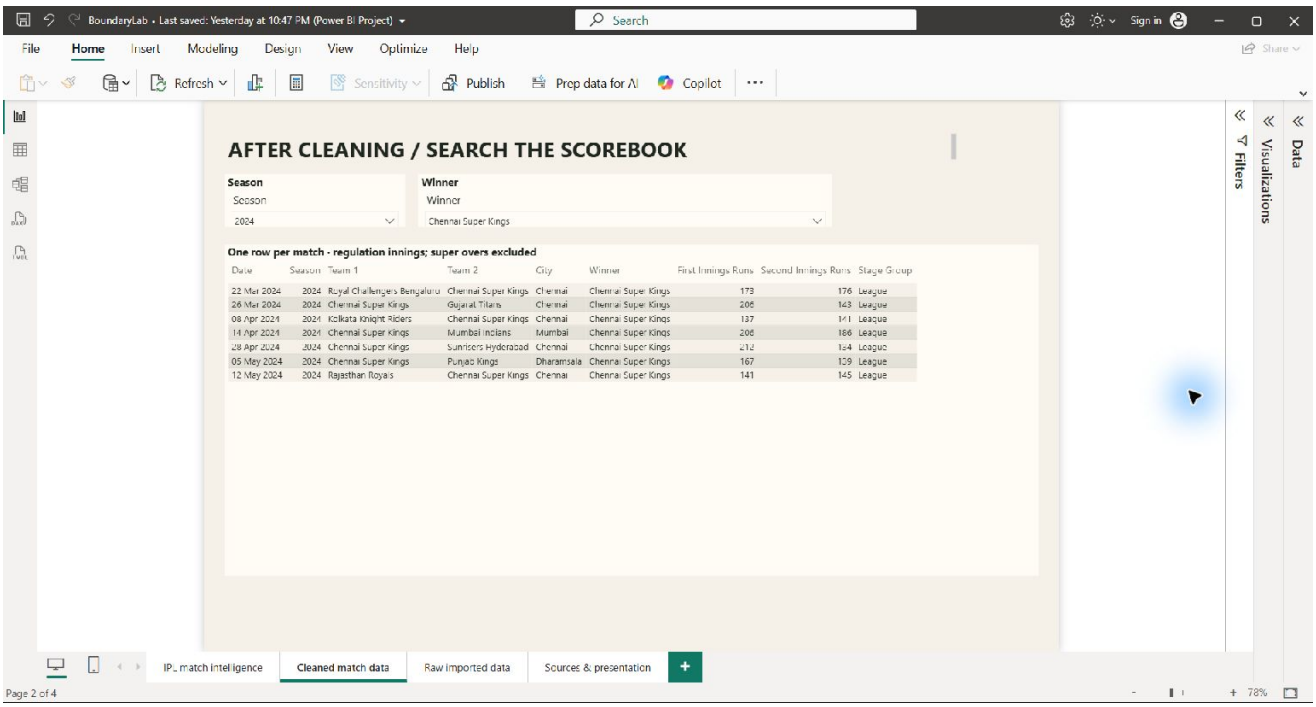
Selecting Season 2024 and Chennai Super Kings narrows the match set to 14 records. The filter context updates the dashboard without deleting or altering the source table.



Dashboard view after the Season 2024 and team selections.

Filtering results and interpretation

FILTERED VIEW	EXPECTED RESULT	READING THE RESULT
Season 2024 + Chennai Super Kings	14 matches	The view narrows from 459 matches to the selected team-season subset.
Chasing wins in the subset	50.0%	Descriptive share within the selected 14 matches.
Toss conversion in the subset	42.9%	Descriptive rate; it does not show that the toss caused a result.
Average first-innings runs	185.4	Subset average; interpret alongside venue and match conditions.
Winner = Chennai Super Kings, Season 2024	7 wins	Filtered winning-match count.

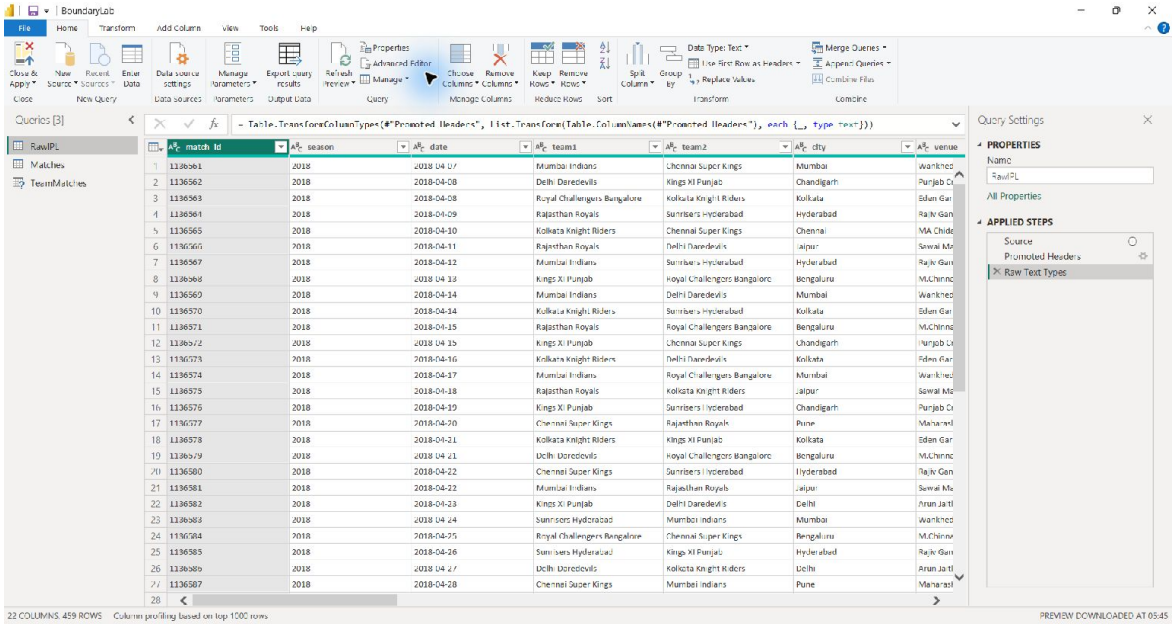


Filtered rows visible in Power BI after applying the demonstration selection.

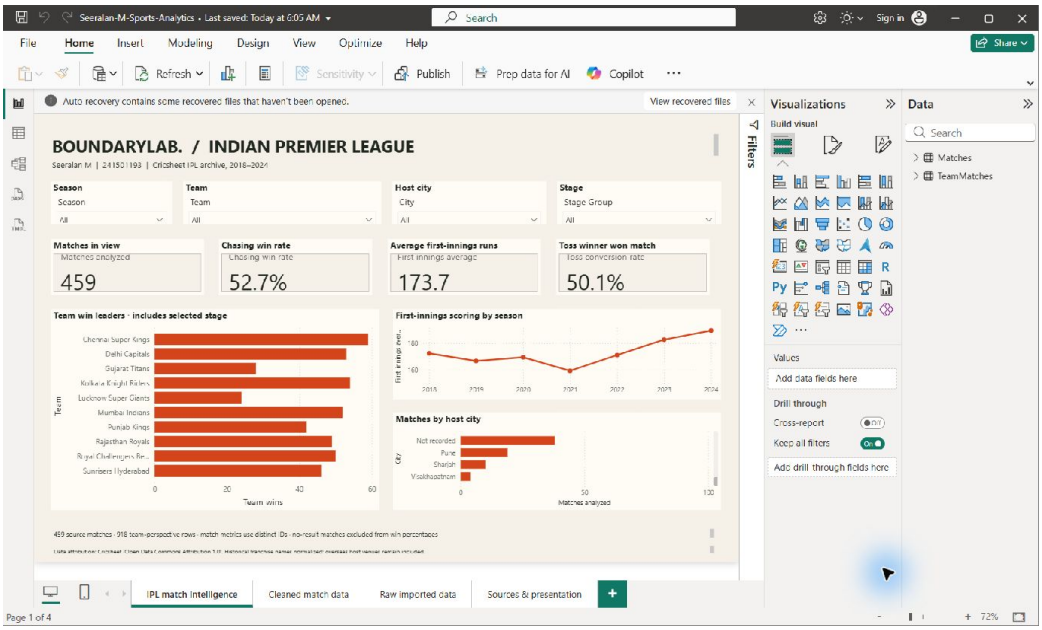
The match-level and team-perspective tables have different grains. Distinct Match ID measures are used for match totals so derived team rows do not double-count matches.

Transformation evidence and portability

The screenshots document the raw query, transformation steps, resulting cleaned query, and model output. The PBIX uses embedded compressed source CSV bytes in its Power Query inputs, so opening and refreshing the project does not rely on a path from the original computer.



Raw import and initial query view.



Refresh and portability evidence from the project capture.

Conclusion and limitations

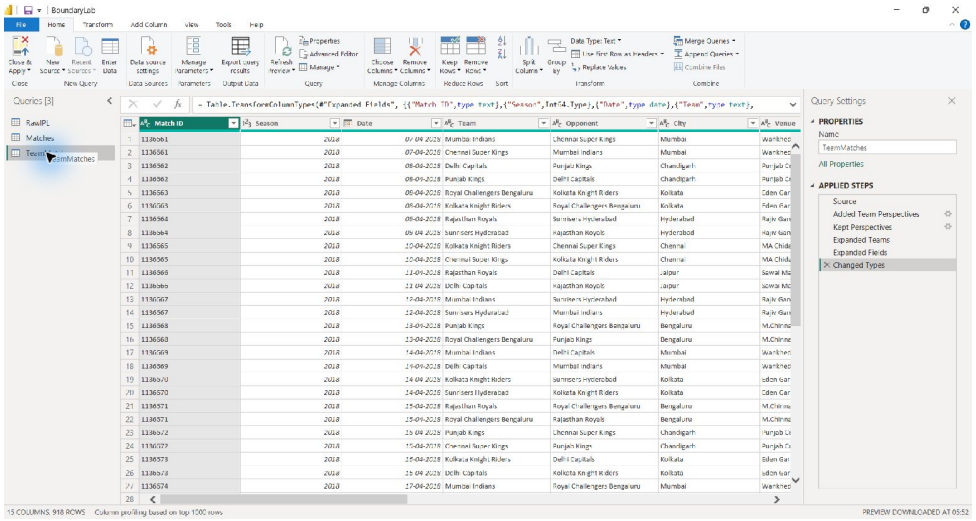
BoundaryLab provides a reproducible, offline workflow for inspecting the available IPL match snapshot, following its data preparation, and comparing full-sample and filtered results in Power BI Desktop. Its main value is transparent descriptive analysis and visual exploration.

Limitations

The application covers only records available in the bundled Cricsheet snapshot. Match totals depend on the selected seasons, stages, venues, and filters. Outcomes are descriptive: chasing, toss, venue, and team comparisons do not establish causal effects or predict future matches. Shortened matches, no-results, and changing franchise identities require interpretation in the context of the recorded data.

Demonstration sequence

1. Open the PBIX in Power BI Desktop and show the overview with slicers cleared.
2. Open the raw imported data page and point out original source fields.
3. Open Transform data and compare the raw query with the cleaned query and Applied Steps.
4. Return to the overview, select Season 2024 and Chennai Super Kings, and explain the changed visuals and 14-match sample.
5. Open the filtered-data page to show the records behind the summary.



Derived team-perspective table evidence; two team rows are represented per match.

References

Cricsheet data downloads | Bundled source archive, prepared CSV, data dictionary, license, provenance record, and screenshots.

End of report